

PASSON

VNR Student-to-Student Reuse & Knowledge Platform

Complete Project Master Guide

One document for the project team: problem, users, solution, marketplace modules, Looking For, Knowledge Shelf, item lifecycle, trust mechanisms, architecture, database, MVP, implementation plan, demo flow, differentiation, safety, and judge questions.

Core idea: Create a VNR-focused student ecosystem where useful items and practical knowledge can move from one batch to another through selling, exchanging, donating, handing over, and sharing knowledge.

Team rule: Build one convincing end-to-end journey instead of many disconnected features.

Scope boundary: PassOn is designed for the VNR student community. Authentication remains a normal account system; VNR is the campus context, not a special login mechanism.

“Pass it forward.”

1. Executive Summary

PassOn is a campus-focused platform designed around the VNR student community. It helps students pass useful items to other students instead of leaving those items unused, while also giving students a structured way to discover what is available, express what they need, and learn from senior experience.

The platform combines a focused student marketplace with two important community layers: **Looking For**, where a student can post a need, and **Knowledge Shelf**, where seniors can leave useful academic, campus, project, placement, and practical guidance.

The platform should feel like a trusted campus exchange rather than a generic e-commerce website. The main value is relevance: the users, campus context, handover process, and knowledge are all tied to VNR.

2. The Problem

Senior students often finish a semester or graduate with items that are still useful: books, calculators, lab-related resources, project accessories, hostel essentials, electronics, furniture-sized campus items, and other reusable materials. Juniors may simultaneously need those same items but have no focused VNR channel to find them.

At the same time, valuable senior knowledge is often scattered across personal chats, friend groups, and informal conversations. A junior may know that a senior has useful advice but not know where to find it.

Core problem statement: How can VNR create a trusted, campus-specific system that helps useful physical resources and practical student knowledge move efficiently from one batch to another?

3. Why Existing Processes Leave a Gap

- Generic resale platforms are not designed around VNR's campus context.
- WhatsApp and other chats are difficult to search and quickly become fragmented.
- A junior may not know which senior has a needed item or useful experience.
- A senior may have useful items but too little time to individually find interested juniors.
- Knowledge shared informally is easy to lose after a batch graduates.

4. The PassOn Solution

One-line pitch: “PassOn connects VNR students so useful items can be reused and useful knowledge can be passed forward.”

The platform is designed around three connected experiences:

Experience	Purpose
Student / Senior	List an item, choose sell/exchange/donate/hand-over, respond to interested students, coordinate handover, and share knowledge.
Student / Junior	Discover items, search by need, post Looking For requests, express interest, and access useful knowledge.
VNR Community	Keep reusable resources and student-to-student knowledge discoverable inside the campus context.

5. Complete End-to-End Architecture

→ Browse / Search → Item or Looking For → Match → Interest / Request → Owner Response → Handover → Completion → Feedback

Knowledge path: Student contribution → Knowledge Shelf → Search / Browse → Student reads → Save / useful feedback.

A simple mental model for the team:

- Resource → Discovery → Match → Handover → Reuse
- Experience → Knowledge Shelf → Discovery → Learning → Continuity

6. User 1 — Senior Student

- Create an item listing with photos, title, category, condition, and exchange mode.
- Choose: Sell, Exchange, Donate, or Hand Over.
- Respond to interested students and coordinate handover.
- Mark an item as reserved, handover planned, or completed.
- Publish useful experience or guidance to the Knowledge Shelf.

User Login: Sign in with a standard account.

7. User 2 — Junior Student

- Browse and search VNR-listed items.
- Filter by category, condition, exchange mode, and relevance.
- Open item details and express interest.
- Post a Looking For request when the required item is not currently listed.
- Receive / discover relevant matches.
- Access the Knowledge Shelf for senior guidance.

8. User Authentication & Profiles

Users sign in with a standard account. Their profile stores basic branch, batch, and campus context used by the marketplace. No special VNR institutional authentication is required for the MVP.

Access layer	Purpose
User account	Provide account-based access to the platform.
Student profile	Show basic batch/branch or relevant campus context without exposing unnecessary personal data.
Role/context	Support senior and junior experiences while keeping the product focused on VNR.
Visibility controls	Avoid unnecessarily exposing contact information or sensitive profile details.

9. Item Listing Module

A student should be able to create a useful listing without filling a long form.

Field	Example
Item title	Engineering Drawing Set
Category	Academic / Study
Condition	Good
Exchange mode	Sell / Exchange / Donate / Hand Over
Expected value	Optional amount or exchange preference
Description	Short condition and usage details
Images	One or more item photographs
Availability	Available / Reserved / Completed

10. Item Discovery

- Search by item name or keyword.
- Browse by category.
- Filter by exchange mode.
- Filter by condition or availability.
- Prioritize relevant VNR listings rather than generic marketplace content.
- Open a detail page before expressing interest.

11. Looking For

Looking For is a first-class module. Instead of waiting for a student to list something, a student can describe what they need.

Input	Purpose
Need title	What the student is looking for
Category	Academic, hostel, electronics, project, etc.
Description	Important requirements or acceptable alternatives
Preferred exchange mode	Buy / Exchange / Donate / Any
Status	Open / Matched / Closed

Example: “Looking for a used scientific calculator for next semester.” A matching listing can satisfy the request, or another VNR student can respond.

12. Matching & Interest Flow

The matching layer connects item listings and Looking For posts using simple attributes such as category, keywords, exchange mode, and availability. For the prototype, transparent rule-based matching is preferable to an unnecessarily complex AI system.

Listing → Match candidate → Student expresses interest → Owner accepts / declines → Handover coordination → Completed

13. Handover / Transaction Lifecycle

Status	Meaning
Available	Item is open for interest.
Interested	A student has expressed interest.
Reserved	Owner has temporarily reserved it.
Handover planned	Both sides have agreed on a handover arrangement.
Completed	Handover is marked complete.
Closed	Listing is no longer active.

The prototype should avoid becoming a full payment platform. The important demonstration is discovery, matching, communication/interest, and successful campus handover.

14. Knowledge Shelf

- Senior tips and lessons learned.
- Course and subject guidance.
- Project and hackathon experience.
- Placement preparation experiences.
- Campus / hostel practical tips.
- Useful resources and explanations.

15. Knowledge Shelf — Content Structure

Content	Example
Title	How I prepared for placement coding rounds
Category	Placement
Author context	Senior student / batch context
Content	Experience, tips, mistakes, resources
Tags	DSA, interviews, projects
Useful feedback	Optional student feedback signal

16. Search & Discovery

- Search items and knowledge posts from one VNR-focused interface.
- Use categories and tags to reduce browsing effort.
- Show clear item status so students do not waste time on unavailable listings.
- Keep Knowledge Shelf searchable after the original contributor has moved on.

17. Senior Dashboard

Dashboard section	Purpose
My Listings	View and manage active items.
Interested Students	See students who expressed interest.
Handover Status	Track reserved / planned / completed items.
My Looking For	See personal requests if applicable.
Knowledge Contributions	Manage Knowledge Shelf posts.

18. Junior Dashboard

Dashboard section	Purpose
Saved / Interested	Track useful listings.
Looking For	Create and manage requests.
Matches	See relevant listings or responses.
Knowledge Shelf	Access saved or recently viewed guidance.

19. Listing Detail View

Section	Information
Item	Title, images, category, condition
Exchange	Sell / Exchange / Donate / Hand Over
Description	Owner-provided details
Availability	Current status
Owner context	VNR student context shown with privacy controls

Section	Information
Action	Express interest / request / save

20. Trust & Safety

Trust is especially important in a student-to-student exchange. The platform should make interactions transparent without pretending to provide the protections of a commercial marketplace.

- Do not expose unnecessary personal information.
- Allow users to report inappropriate listings or behavior.
- Show listing status clearly.
- Keep a basic record of listing and status changes.
- Use neutral wording for disputes and avoid automatic accusations.
- Encourage safe campus handover practices.

21. Feedback / Reputation Layer

A lightweight feedback mechanism can help students understand whether completed exchanges were smooth. For the MVP, keep this simple rather than building a large gamification system.

Signal	Use
Completed exchange	Confirms lifecycle completion.
Basic feedback	Helps identify good or problematic interactions.
Report	Allows a user to flag a concern for review.

22. Admin / Moderator Module

- Review reported listings or users.
- Remove clearly inappropriate content according to platform rules.
- View basic platform activity.
- Handle disputes or escalations manually.
- Manage categories and basic content settings.

23. Complete User Flow

Login → Home → Browse / Search OR Looking For → Item Detail → Express Interest → Owner Response → Reservation → Handover → Completed → Feedback

Knowledge flow: Knowledge Shelf → Search / Category → Read → Save / Useful feedback.

Principle: Keep the platform focused on the intended VNR community.

24. Complete Technical Architecture

Layer	Suggested technology
Frontend	React / Next.js + Tailwind CSS
Backend	Node.js / Express or a suitable API layer
Database	PostgreSQL / Supabase
Authentication	Email/password or Google login
Storage	Object storage for item images
Search	Database search initially; richer search later if needed
Hosting	Suitable frontend + backend hosting

The exact stack should follow what the team can build reliably. The goal is a convincing PassOn prototype, not an impressive-looking technology list.

25. System Architecture — One Picture in Words

VNR Student



Authentication & Profile



Marketplace / Looking For / Knowledge Shelf



Listings + Requests + Interests + Knowledge



Matching + Status Lifecycle + Moderation



Database / Storage

26. Core Database

Table	Core fields
users	id, name, email, branch, batch, role, created_at
listings	id, owner_id, title, category, condition, mode, description, status, created_at
listing_images	id, listing_id, file_url, created_at
interests	id, listing_id, student_id, status, created_at
looking_for	id, student_id, title, category, description, mode, status
matches	id, listing_id, request_id, match_score/rule, status
knowledge_posts	id, author_id, title, category, content, tags, status
reports	id, reporter_id, target_type, target_id, reason, status
feedback	id, exchange_id, from_user, rating/feedback, created_at

27. Listing & Request Rules

- Only authenticated users can create or interact with marketplace content.
- Every listing has a clear availability state.
- A completed listing should no longer appear as actively available.
- Looking For requests can be closed once matched or no longer needed.
- Images should be associated with the listing record.
- Content edits should preserve the current state and basic timestamps.

28. Matching Logic — Prototype

A transparent prototype can calculate a simple relevance score using matching attributes. This is an implementation recommendation, not a claim that complex AI is necessary.

Signal	Example contribution
Category match	Strong match when categories align.
Keyword overlap	Higher relevance when title/description terms overlap.
Exchange mode	Prioritize compatible buy/exchange/donate preferences.
Availability	Active listings rank above unavailable ones.
Campus relevance	All results remain within the VNR-focused ecosystem.

Example: Looking For = “scientific calculator” + Category = Academic. A listing with the same category and relevant keywords becomes a high-confidence match.

29. MVP — What We Actually Build

Feature	Priority
User authentication	MUST
Item listing	MUST
Item discovery/search	MUST
Sell / Exchange / Donate / Hand Over	MUST
Express interest	MUST
Listing status lifecycle	MUST
Looking For	MUST
Knowledge Shelf	MUST
Basic moderation/reporting	SHOULD
Basic feedback	SHOULD
Complex recommendation AI	FUTURE
Integrated payments	FUTURE / OUT OF MVP

30. Implementation Plan

Phase	Build
1 — Foundation	Frontend structure, standard authentication, database schema
2 — Marketplace	Listing creation, images, categories, search
3 — Interest	Express interest, owner response, reservation
4 — Lifecycle	Handover and completed status flow
5 — Looking For	Need posts and simple matching
6 — Knowledge	Knowledge Shelf, categories, search
7 — Trust	Reports, moderation, basic feedback
8 — Integration	Connect senior → junior → handover → completion
9 — Polish	UI consistency, testing, demo data, presentation

Development principle: Get one complete exchange working from listing to handover before adding advanced features.

31. Team Division

Role	Responsibility
Frontend	Senior/junior screens, listing, discovery, dashboards
Backend	APIs, database, authentication, listing lifecycle
Marketplace logic	Search, Looking For, matching and status logic
Knowledge / Content	Knowledge Shelf, categories, moderation flow
UI / Demo	Design consistency, demo story, testing and presentation

32. Demo Data Strategy

Use clearly labeled synthetic/demo listings for the presentation. The dataset should represent different VNR-relevant categories and show both supply and demand.

- Student A lists a used scientific calculator for exchange.
- Student B donates a set of reference books.
- Student C posts a Looking For request for a calculator.
- The platform surfaces a relevant match.
- The student expresses interest and the owner reserves the item.
- The exchange is marked completed.
- A Knowledge Shelf post provides placement/project guidance.

33. 3-Minute Winning Demo

Time	Demo
0:00–0:25	Student logs in and lands on the PassOn campus marketplace.
0:25–0:55	Student opens Add Listing and creates a useful item listing.
0:55–1:20	Another student searches and finds the item.
1:20–1:40	Show Looking For request and matching result.
1:40–2:00	Student expresses interest; owner reserves the item.
2:00–2:20	Show handover and Completed status.
2:20–2:45	Open Knowledge Shelf and show a useful senior post.
2:45–3:00	Finish with the PassOn one-line pitch.

34. Exact Demo Story

Senior: “I have a scientific calculator I no longer need.”

Listing: Academic → Good condition → Exchange.

Junior: Searches for calculator and sees the listing.

Looking For: “Need a scientific calculator for next semester.”

Match: Relevant listing surfaced.

Interest: Junior requests the item.

Senior: Reserves it for handover.

Completion: Item marked handed over / completed.

Knowledge Shelf: Junior then opens a senior-written placement guide.

35. What Makes PassOn Different?

Generic approach	PassOn
Public/general marketplace	VNR-focused campus ecosystem
Individual resale listing	Campus-context exchange
Search only	Search + Looking For
One-time transactions	Reuse + knowledge continuity
Knowledge lost in chats	Persistent Knowledge Shelf
Generic user identity	Student branch/batch context

36. Startup / Campus Potential

The strongest near-term value is campus utility rather than broad consumer commerce. The concept can later evolve into a reusable campus resource layer while retaining VNR as the initial controlled environment.

Area	Possible value
Students	Lower-cost access to useful items and practical knowledge.
Seniors	Easy way to pass on items and experience.
Juniors	Better discovery of resources and senior guidance.
Campus community	Higher reuse and less resource wastage.

37. Trust, Privacy & Safety

- Keep the product focused on the VNR community as a core scope constraint.
- Collect only information required for account, listings, requests, and safe platform operation.
- Do not expose personal contact details unnecessarily.
- Use role-based or authenticated access to user information.
- Provide a reporting mechanism for inappropriate listings or behavior.
- Clearly label demo data during presentations.
- Do not claim the platform guarantees transaction safety or dispute resolution.

38. Judge Questions & Strong Answers

Question	Answer
Why not use a generic marketplace?	Because the core value is VNR-specific relevance, trust, batch context, campus handover, and senior knowledge.
What is innovative?	The combination of a VNR-focused reuse marketplace, Looking For demand posts, and a persistent Knowledge Shelf.
Why Looking For?	It lets students express demand even when the required item has not yet been listed.
Why Knowledge Shelf?	Useful senior experience should remain searchable instead of disappearing into private chats.
Are you building payments?	Not in the MVP. The prototype focuses on discovery, interest, matching, handover and completion.

39. What NOT to Build

- A generalized public marketplace.
- Food delivery or ordering.
- Large-scale payment infrastructure.
- Social-media-style feed and unrelated engagement features.
- Unnecessary gamification or reward systems.
- A large ERP for the college.
- Complex AI recommendations before basic matching works.
- Twenty different dashboards.
- Features that do not strengthen the VNR student-to-student exchange.

40. Final Architecture — One Picture in Words

VNR STUDENT



AUTHENTICATION & PROFILE



Marketplace / Looking For / Knowledge Shelf



Listing + Request + Matching + Interest



Reservation + Handover



Completed Exchange + Feedback



Knowledge remains available for future juniors

41. Final Project Statement

PassOn is a VNR-focused student platform that helps students pass useful items to one another through selling, exchange, donation, or handover, while giving students a Looking For mechanism to express unmet needs and a Knowledge Shelf to access practical senior experience.

The project's philosophy is simple: what one VNR student no longer needs can become useful to another VNR student — and what one senior has learned can become a resource for the next batch.

42. Final Pitch

“Every year, useful books, devices, project materials and practical knowledge are left behind or disappear into private chats. PassOn keeps those resources moving within VNR. Students can list, exchange, donate or hand over useful items; others can discover them or post what they are looking for; and the Knowledge Shelf preserves senior experience for the next batch. It is not a generic marketplace — it is a campus exchange built around reuse, relevance and continuity.”

43. Final Build Checklist

- Basic user login works.
- Student can create an item listing.
- Item image upload works.
- Sell / Exchange / Donate / Hand Over options are visible.
- Student can browse and search.
- Student can express interest.
- Listing status changes correctly.
- Looking For post can be created.
- A Looking For request can surface a relevant listing.
- Knowledge Shelf post can be created and viewed.
- Search / categories work for Knowledge Shelf.
- Basic report/moderation flow exists.
- Demo data is clearly labeled.
- The complete student → student → handover journey works without manual database edits.

44. Suggested Screens

Screen	Purpose
Login	Controlled entry to the platform.
Home	Marketplace discovery + Looking For + Knowledge Shelf entry points.
Browse	Search/filter listings.
Listing Detail	View item and express interest.
Add Listing	Student creates an item.
Looking For	Create and browse student needs.
Matches	Show relevant listing/request matches.
My Activity	Track interests, reservations and completed exchanges.
Knowledge Shelf	Browse senior knowledge.
Knowledge Detail	Read and save useful guidance.
Profile	Basic VNR student context.
Admin / Reports	Moderation and issue handling.

45. Evaluation Focus

For a project demonstration, the strongest evaluation points are: clear VNR scope, a real problem, a complete working workflow, simple user interaction, useful Looking For functionality, persistent Knowledge Shelf, and a credible database-backed implementation.

46. Scope & Design Notes

PassOn keeps the structure of a complete project master guide: executive summary, problem, solution, user experiences, connected architecture, modules, technical architecture, database, MVP, implementation plan, demo, differentiation, safety, judge questions and final build checklist.

The core modules are intentionally centered on the established project scope: VNR-focused access, student listings, discovery, Sell/Exchange/Donate/Hand Over, Looking For, and Knowledge Shelf. Authentication is intentionally kept to a normal account system rather than requiring VNR institutional SSO or email verification.

The strongest structural principle is to build one convincing end-to-end journey instead of many disconnected features. That principle is carried throughout this guide.

PASSON

Pass it forward.

Document status: Project planning / prototype guide

Primary scope: VNR-focused student-to-student exchange

Core modules: Marketplace • Looking For • Knowledge Shelf

Recommended MVP: One complete listing-to-handover journey